

Create a **Novel** class with private instance variables, constructor parameters, and accessor methods for “title”, “author name”, and “year published”

Create a **BookStore** class that contains an instance variable for the BookStore’s name. The BookStore class has a **main()** method which creates a BookStore object. The BookStore contains a second instance variable that is an List of Novel references; the BookStore constructor populates that ArrayList with the following data:

Title	Author	Year Published
The Adventures of Augie March	Saul Bellow	1953
All the King’s Men	Robert Penn Warren	1946
American Pastoral	Philip Roth	1997
An American Tragedy	Theodore Dreiser	1925
Animal Farm	George Orwell	1946
Appointment in Samarra	John O'Hara	1934
Are You There God? It's Me, Margaret.	Judy Blume	1970
The Assistant	Bernard Malamud	1957
At Swim-Two-Birds	Flann O'Brien	1938
Atonement	Ian McEwan	2002
Beloved	Toni Morrison	1987
The Berlin Stories	Christopher Isherwood	1946
The Big Sleep	Raymond Chandler	1939
The Blind Assassin	Margaret Atwood	2000
Blood Meridian	Cormac McCarthy	1986
Brideshead Revisited	Evelyn Waugh	1946
The Bridge of San Luis Rey	Thornton Wilder	1927
Call It Sleep	Henry Roth	1935
Catch-22	Joseph Heller	1961
The Catcher in the Rye	J.D. Salinger	1951
A Clockwork Orange	Anthony Burgess	1963
The Confessions of Nat Turner	William Styron	1967
The Corrections	Jonathan Franzen	2001

The Crying of Lot 49	Thomas Pynchon	1966
A Dance to the Music of Time	Anthony Powell	1951
The Day of the Locust	Nathanael West	1939
Death Comes for the Archbishop	Willa Cather	1927
A Death in the Family	James Agee	1958
The Death of the Heart	Elizabeth Bowen	1958
Deliverance	James Dickey	1970
Dog Soldiers	Robert Stone	1974
Falconer	John Cheever	1977
The French Lieutenant's Woman	John Fowles	1969
The Golden Notebook	Doris Lessing	1962
Go Tell It on the Mountain	James Baldwin	1953
Gone with the Wind	Margaret Mitchell	1936
The Grapes of Wrath	John Steinbeck	1939
Gravity's Rainbow	Thomas Pynchon	1973
The Great Gatsby	F. Scott Fitzgerald	1925
A Handful of Dust	Evelyn Waugh	1934
The Heart Is a Lonely Hunter	Carson McCullers	1940
The Heart of the Matter	Graham Greene	1948
Herzog	Saul Bellow	1964
Housekeeping	Marilynne Robinson	1981
A House for Mr. Biswas	V.S. Naipaul	1962
I, Claudius	Robert Graves	1934
Infinite Jest	David Foster Wallace	1996
Invisible Man	Ralph Ellison	1952
Light in August	William Faulkner	1932
The Lion, The Witch and the Wardrobe	C.S. Lewis	1950
Lolita	Vladimir Nabokov	1955
Lord of the Flies	William Golding	1954

The Lord of the Rings	J.R.R. Tolkien	1954
Loving	Henry Green	1945
Lucky Jim	Kingsley Amis	1954
The Man Who Loved Children	Christina Stead	1940
Midnight's Children	Salman Rushdie	1981
Money	Martin Amis	1984
The Moviegoer	Walker Percy	1961
Mrs. Dalloway	Virginia Woolf	1925
Naked Lunch	William Burroughs	1959
Native Son	Richard Wright	1940
Neuromancer	William Gibson	1984
Never Let Me Go	Kazuo Ishiguro	2005
1984	George Orwell	1948
On the Road	Jack Kerouac	1957
One Flew Over the Cuckoo's Nest	Ken Kesey	1962
The Painted Bird	Jerzy Kosinski	1965
Pale Fire	Vladimir Nabokov	1962
A Passage to India	E.M. Forster	1924
Play It as It Lays	Joan Didion	1970
Portnoy's Complaint	Philip Roth	1969
Possession	A.S. Byatt	1990
The Power and the Glory	Graham Greene	1939
The Prime of Miss Jean Brodie	Muriel Spark	1961
Rabbit, Run	John Updike	1960
Ragtime	E.L. Doctorow	1975
The Recognitions	William Gaddis	1955
Red Harvest	Dashiell Hammett	1929
Revolutionary Road	Richard Yates	1961
The Sheltering Sky	Paul Bowles	1949
Slaughterhouse-Five	Kurt Vonnegut	1969

Snow Crash	Neal Stephenson	1992
The Sot-Weed Factor	John Barth	1960
The Sound and the Fury	William Faulkner	1929
The Sportswriter	Richard Ford	1986
The Spy Who Came in from the Cold	John le Carré	1964
The Sun Also Rises	Ernest Hemingway	1926
Their Eyes Were Watching God	Zora Neale Hurston	1937
Things Fall Apart	Chinua Achebe	1959
To Kill a Mockingbird	Harper Lee	1960
To the Lighthouse	Virginia Woolf	1929
Tropic of Cancer	Henry Miller	1934
Ubik	Philip K. Dick	1969
Under the Net	Iris Murdoch	1954
Under the Volcano	Malcolm Lowry	1947
Watchmen	Alan Moore and Dave Gibbons	1986
White Noise	Don DeLillo	1985
White Teeth	Zadie Smith	2000
Wide Sargasso Sea	Jean Rhys	1966

Create the following BookStore methods using appropriate methods:

Method name	Parameters	Resulting behavior
printAllTitles	none	Prints all titles in UPPERCASE
printBookTitle	String title	Prints all titles that contain the specified parameter
printTitlesInAlphaOrder	none	Prints all titles in alphabetical order, A-Z
printGroupByDecade	int decade	Prints all books for the inputted decade IE 2000s -> print all book titles from 2000 - 2009
getLongest	none	Find the longest title in the bookstore
isThereABookWrittenIn	int year	Returns true or false
howManyBooksContain	String word	Returns the amount of Books that contain this word in their title
whichPercentWrittenBetween	int first, int last	What percentage of the books were written between these two years (inclusive)?
getOldestBook	none	Returns the oldest book
getBooksThisLength	int titleLength	Returns a List of all books whose title is this length
main	String[] args	Calls all of the BookStore methods listed above, in this table

Your main() method will look like this:

```
public static void main(final String[] args) {  
    final BookStore bookstore;  
    final Novel oldest;  
    final List<Novel> fifteenCharTitles;  
  
    bookstore = new BookStore("Classic Novels Collection");  
    System.out.println("All Titles in UPPERCASE:");  
    bookstore.printAllTitles();  
    System.out.println("\nBook Titles Containing 'the':");  
    bookstore.printBookTitle("the");  
    System.out.println("\nAll Titles in Alphabetical Order:");  
    bookstore.printTitlesInAlphaOrder();  
    System.out.println("\nBooks from the 2000s:");  
    bookstore.printGroupByDecade(2000);  
    System.out.println("\nLongest Book Title:");  
    bookstore.getLongest();  
    System.out.println("\nIs there a book written in 1950?");  
    System.out.println(bookstore.isThereABookWrittenBetween(1950));  
    System.out.println("\nHow many books contain 'heart'?");  
    System.out.println(bookstore.howManyBooksContain("heart"));  
    System.out.println("\nPercentage of books written between 1940 and 1950:");  
    System.out.println(bookstore.whichPercentWrittenBetween(1940, 1950) + "%");  
    System.out.println("\nOldest book:");  
    oldest = bookstore.getOldestBook();  
    System.out.println(oldest.getTitle() + " by " + oldest.getAuthorName() + ", " + oldest.getYearPublished());  
    System.out.println("\nBooks with titles 15 characters long:");  
    fifteenCharTitles = bookstore.getBooksThisLength(15);  
    fifteenCharTitles.forEach(novel -> System.out.println(novel.getTitle()));  
}
```

...and the output will look like this...

BookStore: Classic Novels Collection

All Titles in UPPERCASE:

THE ADVENTURES OF AUGIE MARCH

ALL THE KING'S MEN

AMERICAN PASTORAL

...

WHITE NOISE

WIDE SARGASSO SEA

Book Titles Containing 'the':

The Adventures of Augie March

The Assistant

...

The Sun Also Rises

Gone with the Wind

All Titles in Alphabetical Order:

1984

A Clockwork Orange

A Dance to the Music of Time

...

White Noise

White Teeth

Wide Sargasso Sea

Books from the 2000s:

Atonement

Never Let Me Go

The Blind Assassin

The Corrections

White Teeth

Longest Book Title:

Are You There God? It's Me, Margaret.

Is there a book written in 1950?

true

How many books contain 'heart'?

3

Percentage of books written between 1940 and 1950:

13%

Oldest book:

A Passage to India by E.M. Forster, 1924

Books with titles 15 characters long:

The Corrections

Light in August

Never Let Me Go

Part 2

In a `Bookshop` class's constructor, create a `Map` of all the Novels; the key is the title and the value is the `Novel`. Then create a set of keys. Use an iterator to loop through the keys (and therefore the map), and print all the titles. Then remove all novels whose title contains "the" in its title (in any letter case). `Novel` overrides `toString()` to print all its instance data in a nice format. Get a set of the keys and sort them as follows to print the novels (this time in sorted order, with no titles containing "the"):

```
// Step 1: Get the keys from the HashMap
final Set<String> keySet = map.keySet();

// Step 2: Convert the Set to a List (because List can be sorted)
final List<String> keyList = new ArrayList<>(keySet);

// Step 3: Sort the List
Collections.sort(keyList);

// Use the sorted keys to get the Map values
```